

CustomMassRanging 4.0

Ranging, Composition, and Multi-Hit Analysis Extension for IVAS

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Overview

The “**CustomMassRanging**” extension provides an automated re-ranging scheme for existing IVAS™ (Integrated Visualization & Analysis Software) ranges. It also provides composition determination based on background correction, as well as multi-hit analysis features for .APT format datasets with the necessary data sections and some ability to estimate losses due to detector pile-up. The motivation for this extension is to provide users with the tools necessary to replicate the analysis reported in a previous publication (Prosa and Oltman, “Study of LEAP® 5000 Deadtime and Precision via Silicon Pre-Sharpened-Microtip™ Standard Specimens”, Microsc. Microanal. 2022).

Standard IVAS features for range and background determination are limited. This extension takes as input a mass spectrum and an initial list of ranges made by a user. Based on a set of user-selected parameters, 1) additional peaks are discovered, 2) range definitions are optimized based on an algorithm that maximizes the net counts for each peak (based on a selected background ranging scheme), and 3) ionic and decomposed compositions are computed after background subtraction with 4) assessment of statistical significance. Should the user choose, the optimized definitions for the original user-identified ranges (or all ranges, user and discovered) can be returned to the main IVAS session. Once “Applied”, the analysis is “Remembered” via the standard IVAS feature “Save Analysis Tree” which saves the parameter settings used by the extension, and these will be re-loaded when a new IVAS session starts from that saved analysis tree (for more details on the Save Analysis Tree, see the IVAS manual for documentation).

In addition, multi-hit analysis features are also available for suitable datasets. A variety of multi-hit tables are computed to allow users to see how various ion events are correlated in space and time. Separation plots, which plot multi-events as a function of their separation distance (as measured in either reconstruction or detector coordinates) are also made available. Normalization of these curves can allow for estimates of ion events missing due to detector deadtime. See the paper mentioned above for a more complete discussion.

Primary Requirements and Enabled Function

- IVAS session with mass spectrum – allows for re-ranging and composition analysis
- ROIs (regions of interest) with created mass spectrum – allows for re-ranging and composition analysis
- .APT format dataset that includes EPOS, pulse, and pulseDelta sections – enables multi-hit processing features (see IVAS documentation for more information regarding .APT sections)

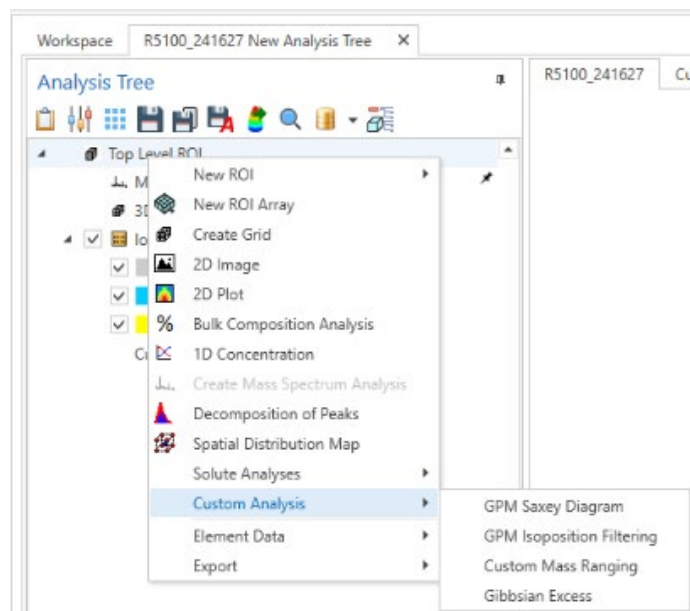
Extension Installation

Ideally, you would go to the Cameca GitHub site: <https://github.com/CamecaAPT>, but this link may not be updated. Alternatively, try <https://github.com/tjprosa41/CustomMassRanging>.

Find the CustomMassAnalysis extension and install (see the AP Suite User Guide, available within a running IVAS session, section 2.8.3 Detailed Instructions for Installing, Updating, or Uninstalling and Extension to AP Suite Using the Extension Ecosystem).

QuickStart

1. Start an IVAS analysis session.
2. Right click hold on any existing ROI.
3. Select “Custom Mass Ranging”.
4. A blank “Custom Mass Ranging” tab will appear with an “Update” button. Click on the “Update” button.
5. The mass spectrum and existing range definitions are loaded and Histogram Information is extracted. Users can modify the default parameters that affect the ranging process in the Parameters section: Ranging, Tail, and Peak Discovery Parameters.
6. Click the “Rerange” button to change the range definitions. (Note: To keep the current/ loaded range definitions, consider checking “Use Manual Ranging” so that reloading this saved analysis tree will not automatically Rerange as a first step).
7. Reranging will discover all peaks by the algorithm, initial ranging schemes have will be determined by the extension, range definitions will be optimized, and compositions will be determined. The user can change parameters and Rerange as many times as desired until an acceptable result has been achieved.



Histogram Information	
Max Peak Name	Si
Max Peak Position (Da)	13.988
Max Peak FWHM (Da)	0.046
Max Peak MRP	304
Spectrum Coarsen Factor	2
Ranging Parameters	
Fixed Ranging Width Factor	2.5
Min Width Factor	0.4
Left Range Criteria	3
Left Range Delta	1.5
Use Fixed Ranging Width	☐
Ignore Discovered Peaks	☒
Rerange Process Multis Help Apply-Remember Export .xlsx	

8. At any time, the user can save all information (all tabs, ranges, and parameters) to a multi-tab Excel file via the “Export .xlsx” button.
9. At any time, should the proper data sections exist in the dataset, the multi-hit tabs “Multi-hit Information” and “Separation Plots” can be populated via the “Process Multis” button. Again, parameters can be changed and “Process Multis” repeated as desired. Note, the key parameter for doing multi-hit analysis is the “Separation Criteria”. It determines which events are considered correlated vs. those which only appear correlated due to statistical coincidence (uncorrelated events).
10. Finally, selecting the “Apply-Remember” button will update the ranging information used in the main IVAS session for this ROI, update the Properties section pertaining to this extension, which makes it savable via the “Saved Analysis Tree” function.
11. Note: Tool tips are available for most variables. Just hover the mouse over an item and a helpful description should appear.

Important Terms

The following are used to help describe the various tracked and computed associations:

- Pulse: Laser or voltage pulse (or both together) attempting to cause field evaporation
Note: Most pulses result in zero detected events
- Event Pulse: One or more ions detected during a single pulse counts as an event pulse
- Multi-hit Event: Two or more ions detected during a single pulse
- Pseudo-double: Evaporation events separated by a small number of pulses
Note: Here, critically defined as consecutive single events only (or events separated by no more than Max dp pulses)
- dp: Pulse difference between detected ions
- Correlated: Events close in both space and time. One event affecting other evaporations
- Uncorrelated: Events occurring independently. Apparent correlations due to random statistics
- Critical Separation: Spatial separation used to partition correlated and uncorrelated events
- Considered: Ion ranges chosen to be tracked for table displays (tracked by ion/peak)
- Other Ranged: Ion ranges defined but not specifically considered (grouped together)
- Unranged: All other ions neither considered nor other (typically, out-of-time events)
- Same-same: Ions from same ion range. dp=0 events of this type are susceptible to pile-up
- Non-same-same: Ion pairs not from same ion range. Events of this type do not pile-up
- hreg: Historical register used to track multi-event type counts: single, double, triple, etc.
- PCME: Probability of Multiple Events (PME) defined as Multi-Pulses/All-Pulses, as used by Cojocaru-Mirédin et al. (Advanced Materials, 2024). Here, correlated-multis are used in numerator instead, so Probability of Correlated Multi Event
- Overcounting: A double (two ions during a pulse) represents a single correlation pair, while a triple (three ions) represents three correlation pairs. Multis are represented in tables this way so that the number-of-ions is generally $\neq 2$ times any table entry

Button: Rerange

Automated Range Definition Determination

The “**CustomMassRanging**” extension provides an automated re-ranging scheme for existing IVAS ranges. Adding or deleting ion ranges must be done in IVAS, before passing them on to this extension. A peak discovery feature displays found peaks and allows the user to add all these ranges (which can be passed back to IVAS for identification and then returned to the extension for computation).

Histogram Information:

Max Peak Name: Ion range of found max counts position

Max Peak Position (Da): Bin position with maximum counts

Max Peak FWHunM (Da): Full width of max peak at hundredth max

Max Peak MRP: Mass resolving power of max peak ($1/\text{FWHunM}$)

Spectrum Coarsen Factor: Binning of the extension mass spectrum (integer number x IVAS binning).

Since IVAS default binning is 0.001 Da, coarsening factor = 2 would result in a 0.002 Da binning.

Range boundaries are determined by maximizing the net counts for a given peak (ranged events minus background estimate). Four ranging schemes exist and are initially populated by the algorithm on a peak-by-peak basis, but subsequent re-ranging allows the user to override the background scheme for any peak. The schemes include: Left, Half, Quarter, and LeftTail.

The LeftTail option attempts to fit the tail of the peak with a decaying exponential fit to unranged regions in the tail. Ionic and decomposed ion bulk compositions use the tail estimate and full statistical consideration is made for detection threshold and resulting compositional uncertainty. When using the LeftTail option, the user specifies an estimated uncertainty in the tail estimate and that error is propagated to the bulk compositions.

Ranging Parameters:

Fixed Ranging Width Factor: Number x FWHunM for left-ranging scheme width (or the fixed ranging width when selected as such)

Min Width Factor: Number x FWHunM for min range width (all ranges must be at least this wide)

Left Range Criteria: Required pre-peak region (Da) empty of other discovered ranges for left-ranging

Left Range Delta: Position left of peak (Da) to center left-background range

Use Fixed Ranging Width: Use fixed ranging width

Ignore Discovered Peaks: Ignore discovered peaks when determining ranges (and compositions)

Schemes

Left (and LeftTail): Designed for peaks with no pre-peaks (flat background with no peak tails overlapping peak region). Uses a fixed range-width, positioned to provide maximum counts. Peak width is set to $\text{RangingWidthFactor} \times \text{FWHunM}$.

Half: First a range of minimum width (even number of bins) is centered about the maximum peak position. Immediately adjacent ranges of $\frac{1}{2}$ width are used to approximate the background. The range width is increased two bins at a time (and each adjacent background range by one bin) with the position moved right or left, depending on which gives the maximum net counts. The range width continues to grow and move two bins at a time until the net counts begin to decrease.

Quarter: Used when peaks are too close together to allow for $\frac{1}{2}$ -range background ranges. The difference is that the immediately adjacent background ranges are $\frac{1}{4}$ width and the range width is increased by four bins at a time.

Note: The background estimates are plotted as straight lines in the Mass Histogram plot as this is the equivalent interpretation for the adjacent ranging method.

LeftTail

There are occasions where one or two dominant peaks both interfere with adjacent peaks and also represent significant numbers of ion events such that any compositional analysis that ignores these will systematically underrepresent these species and provide inaccurate results. The LeftTail feature first determines ranging for such a peak using the Left method described above. The tail region (or region immediately to the right of the peak) is then fit with an exponential curve (straight line in a log plot). Any bins where a range from another peak is defined are ignored. The counts in the fit, extending down to where the tail meets the background, are tallied and used in composition analysis. Any unrealistic tail fits are ignored (e.g., tail extends beyond Tail Range Maximum).

Tail Parameters:

Considered Tail Range (Da): Range beyond right-range-edge to fit exponential tail

Tail Estimate Uncertainty: Estimated fractional uncertainty of tail counts for error propagation

Tail Range Maximum: Fit tail intersection with background maximum width (or error)

Peak Discovery

Any missing peak are “Discovered”, and the user can optionally add all these ranges or ignore them (default). Added ranges become part of the ionic composition but are excluded from the atomic composition calculations (unless the user identifies the ion types in IVAS mass spectrum and then returns/re-imports those ranges to the extension).

Discovery is done via a kernel with a width defined by $\text{RangingWidthFactor} \times \text{FWHM}_{\text{unM}}$. This kernel width is adjusted relative to the Max Peak in time-of-flight (ToF). That means that for the region near the Max Peak, the kernel has this width, but the kernel will be more narrow for low mass-to-charge (equivalent ToF width) and wider for higher mass-to-charge

(equivalent ToF width), and at least some minimum bin width as defined in the parameters. The constant ToF kernel moves across the mass spectrum using $\frac{1}{2}$ -width background ranging to decide if a statistically significant net signal exists for the kernel. If this exceeds a 99% confidence level that a signal is above background (and minimum counts parameter is satisfied), then a peak has been discovered and the range extent defined.

Peak Discovery Parameters:

Sensitivity: Number times FWHunM for left-ranging or fixed ranging kernel width

Min Bin Pairs: Minimum bin width for peak evaluation in coarsened bin pairs

Min Counts for Peak Max Bin: Discovered peaks must have a bin with at least this many counts

*Note: Minimum and Fixed Width Scaling: Generally, minimum and fixed peak widths scale in ToF (or $\sqrt{m/z}$) relative to the Max Peak, so only the Max Peak will have the specific values as set forth in the parameter table. Ranges to the left will be slightly more narrow, and those to the right more broad. Depending on parameter definitions, very low m/z peaks might require near zero-width minimums, so the “Min Bin Pairs” parameter prevents this.

Ranging & Composition Details

The initial ion list and range definitions are the RRNG-type data passed from IVAS, and the range list items cannot be added to or deleted within the extension (the user needs to do that in IVAS and then pass that list to the extension; changes are made in the extension and can then be pass back out to IVAS). The ranging procedure is consistent with the analysis described in Prosa and Oltman, 2022:

Overlapping initial range definitions are not allowed (although IVAS does allow). An exception will be generated and the user notified when input ranges have overlaps.

- 1) The most intense histogram bin in the mass spectrum is discovered and processed for peak m/z position and full-width-hundredth max (FWHunM, note in the publication, the FW0.001M was used, but that metric may is not sufficiently robust for general mass spectra).
- 2) FWHunM is used to determine the histogram m/z resolution. Typically, the input mass histogram has a resolution of 0.001 Da. The algorithm coarsens the resolution so that 15-30 bins encompass the FWHunM of the main peak (typically, somewhere between 0.002 Da to 0.01 Da).

- 3) The peaks are processed from left to right along the mass spectrum to determine the appropriate ranging scheme. Ideally, a background range of the same width as the ranged peak is used to estimate the background for a given peak. The scheme is determined as follows: If there are no peaks defined within a certain distance to the left of a given peak (as defined by “Left Range Criteria”), then the Left ranging scheme will be used. That is, a range defined at some specified distance to the left (as defined by “Left Range Delta”) of the given peak and the same width. If the left-ranging criteria fail, then an evaluation is made for nearest peaks on either side of the given peak. Nearest peak >0.9 Da away is the criteria for a Half-width ranging scheme. Immediately adjacent to the left and right of a prospective peak range, background ranges of half the prospective peak range width are defined and used to estimate the background. The Half scheme is equivalent to estimating a linear background correction for that peak. Otherwise, the Quarter-width ranging scheme is used. Same as the Half scheme, except background counts need to be doubled to account for the range width and provide a linear background correction (and consequently the background uncertainty will be doubled as well).
- 4) Optimum ranges are determined given the initial ion range definitions and auto-determined schemes as follows:
- A minimum range-width is defined based on the FWHunM.
 - The local maximum is determined for each initial ion range.
 - Left-scheme peaks use a fixed range width as specified in Parameters. The position of the fixed-width range is adjusted to yield the maximum number of

Multi	Ion	Peak (Da)	Min (Da)	Max (Da)	Counts	Scheme	Bgd	Tail Count
☒	Si	13.989	13.929	14.133	4,529,637	Left	66	0
☒	Si	14.487	14.403	14.679	230,356	Quarter	4,362	0
☒	Si	14.988	14.889	15.177	150,760	Quarter	956	0
☒	P	15.492	15.372	15.852	14,368	Quarter	956	0
☒	Si	27.975	27.768	28.596	473,692	Quarter	754	0
☒	Si	28.977	28.848	29.088	24,747	Quarter	380	0
☒	SiP	29.493	29.421	29.589	780	Quarter	126	0
☒	Si	29.970	29.847	30.087	16,027	Quarter	214	0

Histogram Information	
Max Peak Name	Si
Max Peak Position (Da)	13.989
Max Peak FWHunM (Da)	0.083
Max Peak MRP	168.5
Spectrum Coarsen Factor	3

Ranging Parameters	
Fixed Ranging Width Factor	2.5
Min Width Factor	0.4
Left Range Criteria	3
Left Range Delta	1.5
Use Fixed Ranging Width	☐
Ignore Discovered Peaks	☒

Tail Parameters	
Considered Tail Range (Da)	1
Tail Estimate Uncertainty	0.05
Tail Range Maximum	5

Peak Discovery Parameters	
Sensitivity	0.5
Min Bin Pairs	6
Min Counts for Peak Max Bin	3

Multi-Hit Parameters	
Key Range	62.0-P2
Separation Criteria	12
Use Nat. Ab.	☐
Use Detector Separations	☐
Use Manual Ranging	☐
Pseudo-Multi Max dp	5
Scaling	Integrated Uncorrelated
Plots List	Multis Only !All
Sep Plots Include	Selected

Rerange

Apply-Remember

Process Multis

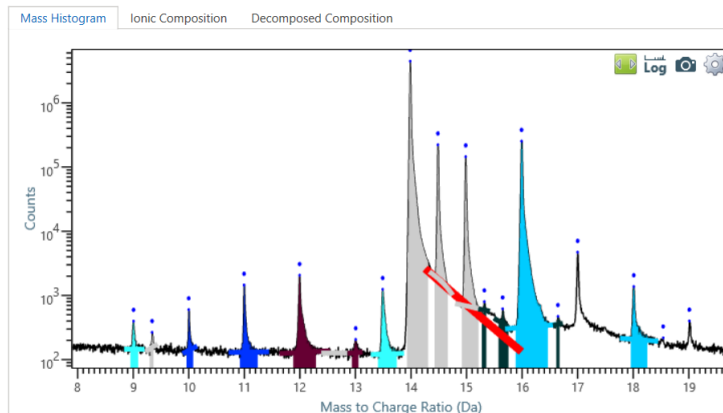
Export .xlsx

Help

total/net counts, and a same-width range is defined to the left of this peak (as specified in Parameters) as a background estimate.

- d. For half-width and quarter-width schemes, the minimum range-width is positioned so that the net counts (total counts in ion range, minus the immediately adjacent background estimate) is maximum. Then the prospective range width is expanded 2 bins (or 4 bins for quarter-width ranging) at a time and re-evaluated. The process stops when the net counts begin to decrease.
- e. Background ranging lines are added to the mass spectrum histogram. Note they are lines in linear-scale but curved in log-scale.

- f. LeftTail performs an exponential fit to the ToF tail ($\sqrt{m/z}$). Starting at the range maximum point and continuing for “Considered Tail Range”, ignoring any values defined by other ranges, these



points are fit to an exponential function and those estimated counts are added to the various tables. Should the exponent be non-negative or span more than 5 Da before intersecting with the background, then the tail is rejected. The user specifies “Tail Estimate Uncertainty” as a fraction of the total number of counts. A value between 0.01 and 0.1 might be reasonable (1% and 10%). This uncertainty is estimated and propagated to the final composition results.

Mass Histogram Ionic Composition Decomposed Composition Multi-hit Information Separation Plots										
Ion	Composition	Sigma/DT (95% CL)	Counts	Backgro	Net	Tail	Missing	Cor. Com	Sigma/DT (95% CL)	
Si	97.3281%	0.0069%	5,425,219	6,732	5,418,613	0	109,043	97.379%	0.039%	
P	1.3678%	0.0051%	78,437	2,298	76,149	0	0	1.3415%	0.0050%	
SiP	0.0516%	0.0012%	3,509	736	2,873	0	0	0.0506%	0.0012%	
P3	0.1023%	0.0014%	5,905	213	5,694	0	0	0.1003%	0.0014%	
P2	1.1134%	0.0045%	63,154	1,168	61,986	0	0	1.0920%	0.0044%	
Ga	0.03682%	0.00085%	2,140	90	2,050	0	0	0.03611%	0.00083%	

Mass Histogram Ionic Composition Decomposed Composition Multi-hit Information Separation Plots										
Element	Composition	Sigma/DT (95% CL)	Counts	Backgr	Net	Tail	Missing	Cor. Comp.	Sigma/DT (95% CL)	
Si	96.0641%	0.0083%	5,428,728	7,468	5,421,486	0	109,043	96.139%	0.046%	
P	3.8996%	0.0084%	225,969	6,009	220,076	0	0	3.8256%	0.0083%	
Ga	0.03632%	0.00083%	2,140	90	2,050	0	0	0.03564%	0.00082%	

- 5) Once the ranging definitions are finalized, ionic and decomposed ion compositions are calculated. For each range (or ion type) the net counts are evaluated for statistical significance (has a peak been detected with a 95% confidence level). Note that any Not Detected (-ND-) ranges are removed from composition considerations and the 95% confidence level detection threshold is reported instead of sigma (the amount of that range/element that would be detected 95% of the time).
- 6) A version of the peak ranging algorithm is used to look for any net counts that statistically might be a peak. The minimum range size is used bin by bin, starting at 0.8 Da. When the net counts are greater than (99%CL-DT/ Sensitivity), this constitutes a discovered peak. The range position is adjusted to yield the maximum net counts for that peak, and then the search process continues starting at the far-right edge of the discovered peak's range extent. Blue dots (limited by plotting options) are displayed on the mass histogram.

Button: Process Multis

Multi-Hit Analysis Tools

Multi-hit analysis tools have been added to this extension, including various tables, separation plots, and statistics computed to help the user assess the evaporation behavior within a particular dataset. Note, the **“GPM.SaxeyDiagram” extension** is also available on the GitHub to display 2D correlation histograms.

Input: Multi-hit analysis can only be performed on **.APT files** that have had additional data sections included at the time of reconstruction. The “Mass” and “Position” data is always included in a reconstruction (POS information). The additional sections include some EPOS sections and two additional, specifically:

Required APT Sections:

- "Position": EPOS:** The reconstructed coordinates (nm)
- "Detector Coordinates": EPOS:** The detector coordinates (mm)
- "Mass": EPOS:** Mass-to-charge ratio (Da)
- "pulse":** Float of pulse number
- "pulseDelta":** Difference between float and double of pulse
- "Voltage": EPOS:** Specimen voltage (no pulse)
- "Epos ToF": EPOS:** Raw Time of Flight minus t0 (ns)

It is noteworthy that EPOS does not include the “pulse” and “pulseDelta” sections, and these would not normally be necessary; however, the assumptions used to decode multi-

hit information from an EPOS file (an entire reconstructed dataset) cannot be used when decoding a sub-volume (region of interest or ROI) from the EPOS array of information. Internally, the added sections enable proper decoding of hit relationships for the extracted lines from the EPOS array:

EPOS sections “Multiplicity” and “Delta Pulse” are defined such that the first ion in a multi-ion list indicates the “Multiplicity” or number of multi-hit events detected for a given pulse, with each succeeding ion in the multi-hit list having “Multiplicity” of zero and “Delta Pulse” of zero, indicating no pulses occurred between events.

This is not reliable for IVAS ROIs because lines of ion information are stripped from the list, leaving this procedure unreliable.

The true pulse number (realPulse) for all ions is decoded internally by:

$\text{double realPulse} = (\text{double})\text{pulse} + (\text{double})\text{pulseDelta};$

Extension Function: After successful “Rerange” one can depress the “Process Multis” button, and as long as all the required sections above exist, multi-hit statistics and ion separation plots (sep plots) will be compiled and displayed in their output tabs (Multi-hit Information and Separation Plot).

The left-most column in the range table section (top of figure to right) allows the user to select ions (referred to below as “selected” ions) to include for specific multi-hit consideration.

Multi-Hit Parameters

The Multi-Hit parameters and functions include:

- 1) Key Range: This isotope-labeled range that is used to track average ToF and Voltage for the analyzed data. The user selects by editing the text before processing, or when no match is found, the first range in the list is used and updated in the parameter list.
- 2) Separation Criteria: The separation distance used to partition the correlated events (those < criteria) and uncorrelated events (those > criteria).
- 3) Use Natural Abundance: Estimated missing counts for same

Multi	Ion	Peak (Da)	Min (Da)	Max (Da)	Counts	Scheme	Bgd	Tail Cour
☒	Si	13.989	13.929	14.133	4,529,637	Left	66	0
☒	Si	14.487	14.403	14.679	230,356	Quarter	4,362	0
☒	Si	14.988	14.889	15.177	150,760	Quarter	956	0
☒	P	15.492	15.372	15.852	14,368	Quarter	956	0
☒	Si	27.975	27.768	28.596	473,692	Quarter	754	0
☒	Si	28.977	28.848	29.088	24,747	Quarter	380	0
☒	SiP	29.493	29.421	29.589	780	Quarter	126	0
☒	Si	29.970	29.847	30.087	16,027	Quarter	214	0
☐	SiP	30.486	30.462	30.510	87	Quarter	72	0
☒	P	30.999	30.780	31.740	64,069	Quarter	1,342	0

Tail Range Maximum	5	☐
Peak Discovery Parameters		
Sensitivity	0.5	☐
Min Bin Pairs	6	☐
Min Counts for Peak Max Bin	3	☐
Multi-Hit Parameters		
Key Range	62.0-P2	☐
Separation Criteria	12	☐
Use Nat. Ab.	☐	☐
Use Detector Separations	☐	☐
Use Manual Ranging	☐	☐
Pseudo-Multi Max dp	5	☐
Scaling	Integrated Uncorrelated	☐
Plots List	Multis Only !All	☐
Sep Plots Include	Selected	☐

Rerange	Process Multis	Help
Apply-Remember	Export .xlsx	

element/molecule pairs that are in the same charge state require knowledge about the relative abundance of these species participating in correlated evaporation. The user can choose to use natural abundance information or an estimate consistent with the acquired data (unchecked).

- 4) Use Detector Separations: Use event separation based on separations in detector coordinates (mm) rather than reconstructed coordinates (nm).
- 5) Use Manual Ranging: This allows users to adjust/override automated ranging upon reloading a saved analysis state. A user can skip the reranging step before Process Multis. The user may then Apply-Remember the data and Save Analysis Tree; however, if the user does not have this box checked, upon updating/loading in the Custom Mass Ranging window when restoring a saved analysis, Rerange will automatically be applied.
- 6) Pseudo-Multi Max dp: Track pseudo-doubles from dp=1 out to Max dp.
- 7) Scaling: Rescale non-same-same separation plots according to selection: None, Max Uncorrelated, or Integrated Uncorrelated. Max Uncorrelated integrates the latter half of the sep distribution histograms (consisting only uncorrelated events). The non-same-same histogram is then rescaled so that it has the same area as the same-same sep histogram for that ion species. Integrated Uncorrelated does the same, but for the range starting with separations defined by the Separation Criteria parameter. Effects of multi-hit deadtime tend to extend beyond the Separation Criteria. The Max Uncorrelated scaling has fewer counting statistics but is less affected by correlated overlaps. Selecting a scaling method allows for estimates of missing counts due to detector deadtime (which are displayed in the legend of the plots).
- 8) Plots List: The Separation Plot tab includes three sets of plots for combinations from each selected ion in the range table section: same-same or pairs of ions from the same defined range (e.g., 14.0-Si – 14.0-Si pairs of events), non-same-same or pairs of ions from differing ranges, and all or all pairs of events regardless of ion range. Since it is all combinations, three selected ion ranges will result in nine plots. Nine more plots for Pseudo-multis. The Plots List selection allows the user to down-select the plots to display: Multis Only, Pseudos Only, Both, Multis Only !All (the same-same and non-same-same histograms, but not the all ions plots, which is the sum of the same-same and non-same-same histograms), Pseudos Only !All, and Both !All.
- 9) Sep Plots Include: Selected, Selected & Other Ranged, All. Discussed later is the idea that multi-hit events can be segmented into correlated and uncorrelated. In addition, specific ion correlations are divided into user-selected ions, all other ranged ions, and unranged ions. For purposes of sep plot display and scaling, the

user can choose to ignore the non-user selected ions and the unranked ions for these calculations and displays.

Multi-Hit Information Tab

This section is likely fluid (will change software version release to version release), as standard terms and descriptive measures are lacking, and perhaps because general software tools are not available APT practitioners. My goal is to provide some tools and present an array of data descriptions in hopes that the potential value of each can be assessed and discussed, eventually leading to some consensus in the APT field (and understanding). I am aware of the works by Oana Cojocaru-Mirédin and David Saxey regarding multi-hit descriptors. Oana defines a Probability of Multiple Events (PME) parameter and suggests that PME (and probability of molecular ions) may be related to the bonding behavior of the analyte. David, besides first reporting the 2D Correlation Histogram (Saxey plot), has presented a Correlation Table where differences between observed ion-pairs and those expected based on their overall ionic abundances were reported in terms of number of standard deviations from expected.

I have included these measures as well as some of my own, where I focus on the correlated events rather than all multi-events measured. For any ion species that has a tendency to evaporate in an overly correlated manner, the differences between including and filtering uncorrelated multi-events may be minor, but for others it may obscure some useful trends.

Simple Multi-Hit Descriptors:

PME: Event probability. The fraction of multi-event-pulses relative to all event pulses.

$$(\text{Number of Multi-Event Pulses}) / (\text{All Pulses with an Event})$$

Ion from PME: Ion origin event probability. Probability that a detected ion originated from any multi-event pulse (approximately $\sim 2 \times \text{PME}$).

$$(\text{Number of Multi Event Ions}) / (\text{Number of Ions})$$

PCME: Ion origin event probability. Probability that a detected ion originated from a correlated multi-event.

$$(\text{Number of Correlated Multi-Events Ions}) / (\text{Number of Ions})$$

PCME Corrected: Ion origin event probability. Probability that a detected ion originated from a correlated multi-event, corrected for estimated events missed due to detector deadtime.

$$(\text{Number of Correlated Multi-Events Ions} + \text{Est. Missing}) / (\text{Number of Ions} + \text{Est. Missing})$$

Deadtime Estimate: Percent of these PCME Corrected events that were not detected due to deadtime.

$$(\text{Missing Ions}) / (\text{Total Ions} + \text{Missing Ions})$$

Corrected Same-Same Efficiency: Correlated same-same events detected relative to total estimated same-same events emitted.

$$(\text{Correlated Same-Same Events}) / (\text{Correlated Same-Same Events} + \text{Missing Ions})$$

Information by Ion Type:

PCME: Probability that this ion type originated from a correlated multi-event.

PCME Corrected: Probability that this ion type originated from a correlated multi-event, adding in estimated missing events.

Missing: Estimated number of missing events for this ion type as a percent of all ions of this type (see later discussion on determination of missing events).

Sigma: The error in the missing percentage as determined by error propagation of counting statistics from the method used.

Pair: Multi-hit ion partner used to estimate missing events. When the pair is the same as the ion type, then the ≥ 3 -isotope method was used.

Singles: Number of single ion events detected for given ion.

Correlated: Number of correlated ions detected for given ion.

Uncorrelated: Number of uncorrelated ions detected for given ion.

Ion Count: Total ions detected for given ion.

Missing: Estimated ions missing due to detector deadtime.

Sigma: Estimated error in estimated ions missing due to detector deadtime.

Raw Composition: Ionic composition based on detected ions.

Corrected Composition: Ionic composition after correcting for estimated deadtime.

Correction Factor: Ratio of raw to deadtime corrected compositions.

Correlated Multis Table, All Multi-Hit Pairs: dpMultis[First Ion, Second Ion, dp=0]

dpMultis[N+2,N+2,DPMax+2] is the internal array that has the tabulated totals for all pairs of multi-events, correlated and uncorrelated. Dp, or delta-pulse, tracks “true” multis (dp=0, or events happening during the same pulse), and “pseudo” multis (dp=1...DPMax, user selects maximum pulse separation to track). This 3D array stores all event pairs and their dp. Ions to track are selected by the user and stored as 0...N-1. N tracks all other ranged ions as a single entry, N+1 tracks any unranged ions, and N+2 stores the totals for the row or column of the array (N+2, N+2 is then the total of the entire table for a give dp, and N+2, N+2, DPMax+2 is the total for all dp>0 events). dpCorMultis[] and dpUncMultis[] track the correlated and uncorrelated multis.

SS Det. Eff. = Same-Same Detection Efficiency or the fraction of same-same ions detected based on the estimated number of missing same-same events.

Note: All combinations of ions are included in these tables/arrays. This means a triple ion event becomes three pairs of events in the appropriate tables/arrays.

Wij Table: Calculated from Ncor and P[i]s based on Correlated Multis Table + 2xMissing.

See later discussion on estimating missing events (see Equation 2 in this draft).

Correlated Multis Table, Deadtime Corrected, #Stdevs:

Saxey Correlation Table: #Stdevs from expected: $(M_{ij} - e_{ij})/\sqrt{e_{ij}}$ where $e_{ij} = 2P_iP_jN$ or P_i^2N where P_i is bulk based and N is the total corrected correlated pairs.

Based on Saxey's correlation table highlighting the statistical deviation, in standard deviations, of the observed number of multi-hit events (in this case, correlated multi-events only, corrected by estimated deadtime) and expected number of multi-hit events based on the ionic composition and total number of observed multi-hit events.

Uncorrelated Multis Table All Multi-Hit Pairs: dpMultis[First Ion, Second Ion, dp=0]

Correlated Pseudo-Multis Table All: dpMultis[First Ion, Second Ion, dp=1...DPMMax]

Uncorrelated Pseudo-Multis Table All: dpMultis[First Ion, Second Ion, dp=1...DPMMax]

Summary:

Total Defined Ranges: Number of defined ranges in ranges table

Key Range: Range defined by user for which to report average ToF and Voltage

Considered Ranges: Ranges selected by user for reporting in table outputs

Separation Critical Value: Key parameter used for defining the separation point between correlated and uncorrelated multi-events

Pseudo-Multi Max dp: Parameter used for tracking maximum pseudo-multis.

Multi Events: Same as PME, percentage of multi-hit pulse relative to all event pulses

Multi Ions: Percentage of multi-hit ions relative to all event pulses

Total Event Pulses

Total Ions

Total Multi Ions

Total Multis Table: Total pairs tracked in dp=1 table, ignoring correlated vs. uncorrelated nature

Multi-dp=0 Distribution Table: singles doubles triples quads higher total

All Events: Number of single, double, etc. events

All Weighted: number of ions represented (i.e., 1*singles, 2*doubles, etc.)

All Weighted: number of ions represented as percent of all ions

Considered Events: subset of events for the considered ranges selected by user

Multis dp=0 Summary:

All: All ions

Considered: Only ions selected by user

Considered & Correlated: Only ions selected by user and correlated

Considered & Uncorr: Only ions selected by user and uncorrelated

Pseudo-Doubles dp=1...DPMMax:

All: All ions

Considered: Only ions selected by user

Considered & Correlated: Only ions selected by user and correlated

Considered & Uncorr: Only ions selected by user and uncorrelated

DR: Average DR

ToF: Average and standard deviation for key ions

Voltage: Average and standard deviation for key ions

Si Isotope Evaluation:

Populated when Si isotopes are present in the selected peaks. Compares corrections based on the method using 3-isotopes (consistent with the paper) vs. 2-isotope method (removed for 4.0 but should be fixed for later versions).

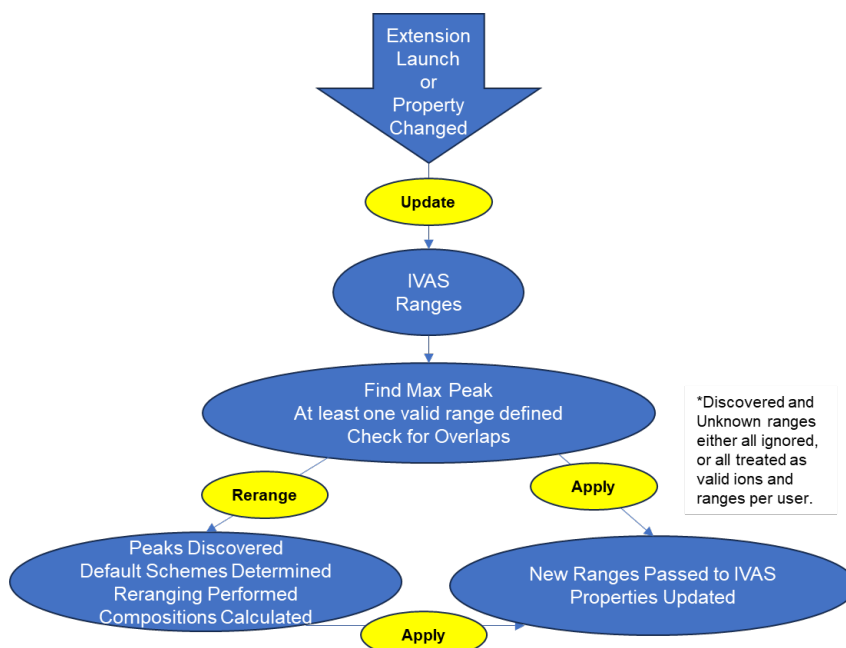
Pseudo-Based Correction Validation (missing counts corrected, but not background corrected): Pseudo multis possess the Pi for correlated events unaffected by deadtime, but may not have sufficient counting statistics. Allows comparison between results for corrections described above the pseudo-multi-based correction (which will be less accurate but should be similar).

Discussion

As discussed in Prosa and Oltman, 2022, atom probe evaporation events do not occur randomly. They present aspects of both correlated evaporation (one evaporation affecting another evaporation) and uncorrelated evaporation (evaporation events occurring independently and randomly in space and time). As a consequence, multi-hit events are tabulated based on time correlations (dp=0 are same-pulse, commonly referenced multi-hit events, and dp=1, or less than some small dp number, as “pseudo”-multis) and based on spatial correlations (events spaced within a small distance likely being correlated while those farther apart following random correlation statistics). Once tabulated, descriptive tables and plots are presented in the “Multi-hit Information” tab for the user to interpret.

Button: Apply-Remember

The user can then choose to pass/apply new ion ranges back to IVAS or change ranging parameters and Rerange again. The utilized extension parameters, those from the last apply-remember operation, are moved to the Properties section so that they can be saved as part of any subsequent “Save Analysis Tree” operation. It also requires “Update” to the extension before continuing.



Button: Help

Pressing the “Help” button should launch a view of the most recent .pdf version of this document.

Button: Export .xlsx

Opens a save window enabling the user to choose name and directory for file save. The Excel file includes tabs documenting all displayed data: Parameters, Ranges Table, Mass Histogram, Ionic Composition, Decomposed Composition, Multi-hit Information, Separation Plots.

Final Notes

1. “Rerange” can be done multiple times (needed when users change Parameters or modify any Schemes).
2. Select/Select-All and Copy/Paste functions should work on all tables (column headers are not captured in this operation). To enable copy/paste in the Multi-hit Information tab, it was made editable. The changes will only temporarily affect the display as any Process Multis operation will restore the display.
3. Double clicking in plot region will unzoom x-axis. A second time is needed to unzoom y-axis. Auto-y-axis scaling is not a current feature.
4. Table columns are also sortable by clicking the column header, but note to control precision display, some numerical columns are of type string, and these will not always sort in numerical order!
5. Quarter-width ranging will have slightly larger uncertainty because the background is scaled by a factor of two (greater background uncertainty).
6. Decomposed compositions error propagation means that any molecular ion with an element that has greater than one atom in the molecular formula (e.g., O in SiO₂) will result in a larger uncertainty because of the scaling of those counts in the decomposition calculation (not historically accounted for in this way in IVAS). This is almost always insignificant, but for materials with a significant fraction of multi-same-atomic molecular species, this could become significant.
7. Tool tips are available for many table column headers, Parameters, and Properties.
8. Any changes in the IVAS ranges associated with an active extension session will force an update of the extension (start from the beginning).
9. Save Analysis Tree saves the current parameters and schemes (those shown in the properties region, which are modified only after apply has been done). Using a Saved Analysis tree will also override auto-scheme (not shown in Properties) with

the previously used scheme for any matching ranges (since a Saved Analysis Tree can be applied to a different dataset than the one from which the save was performed, the IVAS ranges/ions list may be different).

10. “Discovered” range names ignored upon load (user needs to assign new range type/name for them so survive initial load).
11. Ignore function: Rerange always starts with the original StartTable (internal for load)—the original Scheme Properties when they exist, or null. When null, then the scheme will be auto determined.
12. Not Ignore: The peaks list of all found peaks is overridden by any StartTable entries. StartTable scheme is applied if found and null otherwise. When null, then the scheme will be auto determined.
13. The discovery process can provide closely spaced peaks, and the auto-ranging can result in overlapping peaks. The code attempts to resolve these overlaps using the following rules: Same named overlapping ranges, the lower net counts range is removed. Overlapping ranges where one is “Discovered”, the discovered is removed. Any other potential outcomes will cause the extension to fail.
14. For versions beyond 3.0, expect modifications to the multi-hit information tab that may not be captured yet in this document.
15. Send feedback to tjprosa@ua.edu

Postscript: Estimating Missing Counts

Multi-Hit Evaporation and Pile-Up Correction (Draft)

Multi-hit evaporation events can be modeled using basic probability concepts. Consider a population of ions with each ion having a concentration p_i and yielding N pairs of detected events. The number of multi-hit pairs should be proportional to the concentration of each ion in the population. For a homogeneous system of a single chemical species, the expected number of multi-hit pairs, M_{ij} , for ion peaks i and j might be written:

$$M_{ii} = p_i^2 N, M_{ij} = 2p_i p_j N, \sum_j p_i = 1, \sum_{ij} p_i p_j = 1.$$

It is worth noting that for events uncorrelated in space (correlated in time because they occur during the same evaporation pulse, but random in distance—in practical terms, separated by some easily observed critical distance), the number of multi-hit event pairs, N_u , is determined by the detection rate. The number of correlated multi-hit event pairs, N_c , currently has no theoretical basis.

Extending this idea to a homogeneous system with multiple chemical species, one still expects M_{ij} to be proportion to individual ion concentrations, but with weights that depend

in some way on the chemical interactions—that is we know some chemical species generate many more multi-hit events than others. We might describe those systems with weights, w_{ij} , such that:

$$M_{ii} = p_i^2 w_{ii} N_C, M_{ij} = 2p_i p_j w_{ij} N_C, \sum_{ij} p_i p_j w_{ij} = 1.$$

An atom probe delay-line detector (DLD) has been shown to detect multi-hit events very efficiently. To first order, only ions of the same mass-to-charge suffer from detection losses due to pile-up. That is when an ion is detected by the detector, any second ion that hits the detector in approximately the same place and within 3 ns will not be detected while the detector recovers (deadtime). Pile-up has been shown to affect measurement of isotopic abundances and is known to affect composition accuracy for a number of material types (B and P doped silicon, carbides, borides, etc.).

In this white paper, I describe terms and methods to better describe multi-hit behavior and use the statistical patterns to estimate pile-up losses and correct bulk compositions. Previously, I have shown that given three or more isotopes detected with sufficient counting statistics, the pile-up losses can be estimated with no assumptions about the underlying isotopic abundances. Here, I will extend these ideas to systems with only two isotopes and evaluate approaches for correcting single isotope (or single molecular species) peaks.

Three or more peaks of a single chemical species (or molecular species) in the same charge state:

The same chemical species requirement for this category suggests that whatever underlying physics affects the likelihood of a correlated multi-hit event occurring, since the chemical identity of the participants are identical, one would expect the weights to be the same. The same charge-state is important because a previous study showed that the charge-state-ratio (CSR) can be different for correlated and uncorrelated evaporation events. Thus, one cannot assume the weights will be the same for differing charge states nor can one use the uncorrelated CSR to estimate pile-up-free correlated p_i s.

Considering the previous 3-isotope paper, the highest precision for >3-isotope datasets will be achieved by utilizing M_{ij} s for the strongest two isotopes coupled with each of the remaining isotopes. When the most abundant peaks are peaks 1, 2, and 3 in order, then we might model the expected M_{ii} s as:

$$M_{11} = p_1^2 N = \frac{1}{2} M_{12} M_{13} / M_{23} = \frac{1}{2} (2p_1 p_2 N) (2p_1 p_3 N) / (2p_2 p_3 N),$$

$$M_{11} = \frac{1}{2} M_{12} M_{13} / M_{23},$$

$$M_{22} = 1/2 M_{12} M_{23}/M_{13},$$

$$M_{ii} = 1/2 M_{1i} M_{2i}/M_{12},$$

$$\sigma_{Mii} = M_{ii} \sqrt{(1/M_{12} + 1/M_{13} + 1/M_{23})} \text{ for } i = 1 \text{ or } 2,$$

$$\sigma_{Mii} = M_{ii} \sqrt{(1/M_{12} + 1/M_{1i} + 1/M_{2i})}.$$

Example:

Below is a 5.7 million ion region of interest from run R5079_170061. This is a LEAP 5000 XS dataset from a 5 at.% P in Si epi-layer (verified by other characterization techniques). The isotope ratios for the correlated multi-events are far from natural abundance due mainly to pile-up losses from $^{28}\text{Si}^{++}$ - $^{28}\text{Si}^{++}$ pairs. Here we estimate that only 2,043 of the $112,526 \pm 6,269$ expected pairs have been detected. Considering that a total of 22,087 $^{28}\text{Si}^{++}$ were detected as correlated multi-evaporation events, nearly 5X that amount were lost due to pile-up. Note these missing events represent only ~2% of the entire 5.7 million ion dataset.

Multi-Hits	14.0-Si	14.5-Si	15.0-Si
14.0-Si	2,043	10,641	7,360
14.5-Si		14	348
15.0-Si			3
Expected Same-Same			
M11	112,526		
M22		252	
M33			120
Corrected Isotope Frac.			
14.0-Si	0.926		
14.5-Si		0.044	
15.0-Si			0.030

Two peaks of a single chemical species (or molecular species) in the same charge state:

For this case, there is less information not affected by pile-up. In the above situation, the existence of three or more non-same-same M_{ij} s allows for calculation of M_{ii} without using the correlated abundances, p_i . In this case, we have a single M_{ij} and we want two model M_{ii} and M_{jj} —one known, two unknowns (underdetermined). If we can estimate the correlated p s (actually a single unknown since these p s are correlated to each other), then we have two knowns and two unknowns. There are at least two choices for calculating M_{ii} and M_{jj} : 1)

Using the natural isotopic abundances, or 2) using the abundances determined from the pile-up affected dataset. Natural-abundance assumptions will be limited by any fractionation that might be present in the measured sample. Typically, this will be small relative to the correction but may be critical for samples known to be non-natural in abundance and with large sample size. Using the abundances consistent with the detected abundance avoids the natural abundance limitation but will be distorted from the pile-up losses. In many situations, these abundance distortions may have only a small effect on the estimated corrections and might be further overcome by an iterated approach: estimate p,s from raw data, estimate pile-up correction, estimate new p,s from correction and iterate.

Example Natural Abundances and Abundance from Data:

Same dataset as previous. Here, using either the reported natural abundance, or the uncorrected abundances from all the data in the dataset, the expected same-same events are only ~1% different from each other and ~11% lower than the 3-peak method above. Understandably, corrections based on three pieces of information should be more reliable than those made with only two. In terms of error estimates, the natural abundance uncertainties are difficult to determine from the published CIWAA values. For Si, they might be of the order of 3% and 0.3% yielding a 3% uncertainty in the same-same estimates. Using abundances estimated from the data, the precision will go with $1/\sqrt{N}$, and for this dataset is on the order of 1%.

Multi-Hits	14.0-Si	14.5-Si
14.0-Si	2,043	10,641
14.5-Si		14
Natural Abundance		Abundance from Data
14.0-Si	0.9223	0.9235
14.5-Si		0.0467
		0.0460
Expected Same-Same		Expected Same-Same
M11	105,077	106,739
M22		269
		265

Example Abundance from Data, Iterated:

The above example could now fold in the extra 106,739-2,043 and 269-265 missing ions for the 14.0-Si and 14.5-Si respectively, recalculate the abundances, and determine updated expected numbers of same-same events and continue this iterative correction until no there is no change in missing ions and abundance; although, there could be cases where this process does not converge.

Plan to insert this example here from the software later...

Two peaks of a single chemical species (or molecular species) differing in charge state:

Previously, we relied on the fact that the considered peaks were the same chemical species and charge state, so that the weights for each combination were likely the same, and we could utilize reliable information regarding the fraction of each peak participating in the correlated evaporation.

For this case, we still have a single chemical species (single isotope of minimal intensity) with two (or more) charge states, but nothing reliable regarding the fraction of each peak participating in correlated evaporation. It has been shown that the CSRs for correlated evaporation events in Si are different than the uncorrelated events (and we have no theory to relate one to the other), so the uncorrelated CSR cannot be used. Likewise, since there is only a single non-same-same pair to use, we cannot use that to estimate the relative fractions because by definition, there is one of each in the pair (50/50, not useful). One might consider the total number of these ions from all correlated event pairs (that is from pairs that include chemical species or molecules that are not of the same chemical species), but there is no reason to believe that this fraction will be representative of the same chemical species CSR fraction. We seem to be stuck.

Here I'll introduce the term pseudo-correlated multi-hit event. PCOR are pairs of events that occur with separations below some critical value, and only 1 or a few pulses apart. For multi-events occurring randomly, the number of pulses between events, dp , is described by an exponential:

$$f(dp) = DR \exp(-DR*dp).$$

This predicts that there should be more occurrences of pairs of events for small dp (and also a long trail of occurrences with dp much longer than the average DR to balance out the timing so that the average $dp = DR$); however, when one looks at the separation plots, there are clearly a spatially correlated surplus (not predicted) as well as an uncorrelated distribution as predicted. If we assume that these PCOR events obey the same fractional participation and weights as the $dp=0$, COR events, then we can use these pile-up free events to estimate the CSR for correlated events only of the chemical species of interest.

If we let M' represent the PCOR events for $dp=1$ to some small max value, then the expected pile-up affected same-same events and COR-CSR are:

$$M_{11} = \frac{1}{4} M_{12} M_{12}' / M_{22},$$

$$M_{22} = 1/4 M_{12} M_{12}' / M_{11},$$

$$CSR = \sqrt{M_{11}' / M_{22}}.$$